

Big Data with Machine Learning: A Review

Author One^{1*}, Author Two¹, Author Three¹, Author Four¹,
Author Five¹, Author Six¹, Author Seven¹, Author Eight¹

^{1*}Department of Computer Science and Engineering, University, Street,
City, 000000, State, Country.

*Corresponding author(s). E-mail(s): author.one@example.edu;
Contributing authors: author.two@example.edu;
author.three@example.edu; author.four@example.edu;
author.five@example.edu; author.six@example.edu;
author.seven@example.edu; author.eight@example.edu;

Abstract

Big-data machine learning changes quickly, and the 2025–2026 record has not been consolidated. This review examines 20 papers drawn from four journals: IEEE Transactions on Big Data, Big Data Research, Journal of Big Data, and Artificial Intelligence Review. Five papers come from each journal, split between 15 original research articles and five review articles that are kept as a documented exception. Papers were found through programmatic retrieval, a keyword screen, and Crossref verification. We synthesize them thematically rather than paper by paper. The corpus supports one organizing claim: across these journals, big-data machine learning has moved from learning on more data to learning under constraints, and the constraints are privacy, platform, and proof. Scalable methods optimize a resource rather than an accuracy leaderboard. Platforms are described architecturally, and measured latency is rare. Privacy-preserving training competes with scale-maximizing training, and the two programs pull in opposite directions. Evaluation is fragmented, and the corpus’s only reproducibility audit shows that single-run results overstate model stability. Ten of the 20 papers could not be read in full, so several claims are reported at the level the available evidence supports.

Keywords: Machine Learning, Big Data, Literature Review, Privacy, Data Platforms, Evaluation, Reproducibility

1 Introduction

Big-data machine learning sits at the crossing of two fast-moving fields. Data volumes keep rising, and machine learning is the main tool for extracting value from them. The result is a literature that grows faster than any single reader can track. Review articles that consolidate the field are published continuously, yet their coverage is uneven: some survey methods, others survey infrastructure, and few survey both together. A reader who wants to know what the 2025–2026 research actually argues has to assemble that picture from many papers. This review builds such a picture for a defined corpus.

The term “big data” causes trouble before the analysis begins. Outside the literature it usually means volume: data too large for one machine. Inside the corpus reviewed here the word carries a different load. Several application papers use “big data” to describe data variety or image volume rather than distributed scale. One trains thirty thousand images on a single cloud GPU and names no cluster engine. Another processes about twenty-five thousand utterances through an inference runtime on one machine. A third simulates ten clients for a federated healthcare model with no distributed platform at all. This review does not resolve the ambiguity by fiat. It treats the gap between the word and the systems it names as one of its findings.

The corpus is fixed and small by design. It holds 20 papers, five from each of four journals: IEEE Transactions on Big Data, Big Data Research, Journal of Big Data, and Artificial Intelligence Review. Because the assignment requires at least 20 papers from these four journals and caps each at five, the split is exactly five per journal. All items fall in the 2025–2026 window. The review period was set as 2025–2027, but a check against the available record found no 2027-dated item, so the corpus is 2025–2026. Fifteen items are original research articles. Five are review articles, all from Artificial Intelligence Review, which is a review-designated venue. A supervisor clarification recorded that “research papers” means original research articles, and the five review articles are kept as a documented exception. The reviews frame the analysis; every load-bearing number in this paper comes from an original research article.

Selection followed three stages, shown in Fig. 5. Programmatic retrieval from OpenAlex returned 1,519 items published in 2025 or later across the four journals. A keyword screen for a joint machine-learning and big-data signal cut that set to 441. Candidates were verified against Crossref, leaving 49, from which 20 were included. The screen favours joint relevance, so domain papers that mention big data only in the title are underrepresented relative to their true share. The theme balance is curated rather than drawn at random, and that choice shapes what the review can and cannot conclude.

Two limits on the evidence base are stated here rather than buried later. First, the five review articles are synthesis anchors, not primary evidence. Every numerical claim in this review is taken from an original research article’s full text, never from a review’s second-hand table. Second, part of the corpus is inaccessible. Ten of the 20 papers could not be read in full. Seven are described from indexed abstracts only, and three have no indexed abstract at all. Those papers are reported at the level their available evidence supports: descriptions without unverified numbers. Where a value is missing, this review says so instead of supplying one.

The review is organized as follows. Section 2 is the Literature Review, which opens with background and a two-axis taxonomy, then works through scalable methods, platforms and pipelines, applications and domains, cross-cutting concerns and evaluation practice, and future directions. Section 3 is the Tables and Figures component, which narrates the comparison tables and the five figures. Section 4 is the Conclusion.

2 Literature Review

2.1 Background and conceptual scope

Big data is usually described by its “V” properties: volume, velocity, variety, veracity, and value. Volume is the raw size of a dataset. Velocity is the rate at which data arrives, which matters for streams. Variety is the mix of data types, from tables to images to graphs. Veracity concerns data quality and trustworthiness. Value is the usefulness the data yields once analysed. A machine-learning pipeline at scale adds its own stages on top of these properties: ingestion, storage, preprocessing and feature construction, model training, serving, and monitoring. Each stage can become the bottleneck, and which stage binds is an empirical question rather than a fixed one.

The corpus answers that question in a particular way. Its papers treat heterogeneity and constraint as the defining features of big-data machine learning more often than they treat volume. Time-series work dominates the methods branch, graph structure recurs across security, network, and finance domains, and multimodal fusion appears in healthcare and affective computing. The strict reading of “big data” as distributed scale appears in only a few items, and even there it is often an architectural claim rather than a measured one. This review keeps that distinction visible throughout: the taxonomy in the next subsection maps the corpus as it is, not as the popular term implies.

2.2 Taxonomy of big-data machine-learning research

The corpus organizes along two axes, shown in Fig. 2. Axis 1 asks what is learned: the model family and the target task, from classical regression through deep networks to foundation models. Axis 2 asks what surrounds the learning: the platform the model runs on, the domain it serves, and the governance that constrains it. Together the axes define five branches. Branch B1 covers scalable and interpretable learning methods. Branch B2 covers platforms, pipelines, and infrastructure. Branch B3 covers applications and domains. Branch B4 covers foundation models and large language models. Branch B5 covers privacy, governance, and evaluation.

Every paper sits in exactly one branch, and the branch assignment drives the structure of the sections that follow. Table 2 lists the mapping. B1 holds five papers, B2 holds three, B3 holds five, B4 holds four, and B5 holds three. The total is 20, and no paper is counted twice. A few papers straddle branches, and those are marked rather than hidden: federated learning appears as both a privacy mechanism and a 6G infrastructure concern, and one large-language-model method paper runs on industrial production data.

Two cautions apply to the taxonomy. First, it is this review’s classification of this curated corpus, not a measurement of the field’s proportions. The keyword screen guarantees that. Second, its empty cells matter. No paper addresses reinforcement learning at scale, no paper makes a purely theoretical contribution, and no paper carries a 2027 date. Those absences are findings about the corpus and are revisited in Section 2.6.

2.3 Scalable machine-learning methods

Branch B1 answers a narrow question: how do methods scale? In this corpus the answer is plumbing. Four of the five research papers in the branch optimize a resource rather than an accuracy leaderboard. One removes the need to retrain, one trades representation quality for graph geometry, one scales a classical estimator, and one attacks the cost of explaining a model. A sixth item, a review of multimodal data integration, anchors the method arc from classical learning to foundation models that runs through the branch.

The clearest purpose-built transformer is the Knowledge Aggregation Transformer Network, or KATN [1]. It targets multivariate time-series classification and tackles a specific integration problem: dual-network designs exist, with one branch for local features and another for global relations, but merging the two is hard. KATN stacks four aggregation blocks, and each block additively merges a modified residual network for local extraction with a multi-head attention network for global relations. A dense alignment layer and a GELU activation finish each block. The reported protocol is rank-based rather than accuracy-based: across 13 UEA archive datasets, KATN records the lowest average rank against six state-of-the-art transformer variants, with a win-tie-loss record of 9/6/15, and the lowest average rank against 18 existing multivariate time-series classifiers. The evidence form is itself informative. The paper’s headline is a cross-dataset rank aggregate, and the abstract reports no per-dataset accuracy, no variance, and no efficiency cost. That reporting style is exactly what the corpus’s reproducibility audit later finds under-specified, and it links this paper to the evaluation thread in Section 2.6.

Where KATN assumes a static archive, the streaming approach assumes the opposite. Gaussian Space Trees and a Gaussian Weighted ADWIN AutoEncoder both target anomaly detection on streams without offline retraining [2]. The first pairs a space-partitioning ensemble with Gaussian distributional modelling. The second couples an autoencoder with the ADWIN adaptive window, which lets the model react to drift. The two methods are complementary in the reported results: on the ECG5000 dataset, the space-tree method is reported above 94% on all metrics while the autoencoder variant reaches an ROC-AUC above 89%; on the SMTP dataset both stay above 80% ROC-AUC; on credit-card fraud, both are reported at recall above 82% and ROC-AUC above 94%, with low false-positive and false-negative rates. Those values are threshold statements taken from the indexed abstract, not precise measurements, and the abstract names no baselines. The paper is nonetheless the corpus’s cleanest case of a method built for velocity rather than volume: it treats the stream as the setting, and it avoids the offline-retraining cycle that later telecom survey work calls obsolete.

A third route scales through representation geometry. One paper embeds network-traffic graphs in hyperbolic space, using a Poincaré-ball embedding with a novel distance gain factor [3]. The pipeline is unsupervised and has three stages: mutual-information slicing removes redundant features, the Poincaré embedding sets edge weights from a distance function, and a graph attention network classifies anomalies using those weights as initial attention. Deep Graph Infomax supplies graph augmentation. The paper’s central claim is that optimizing edge weights contributes about twice as much as optimizing node attributes to detection quality; on the CICIDS2017 dataset the edge-weight gain is an F1 improvement of 0.1355 against 0.0438 from node augmentation. Across its three datasets, the method records F1 of 0.8366 on CICIDS2017, 0.9341 on UNSW-NB15, and 0.9913 on a proprietary campus log, beating LSTM, GCN, a graph encoder, and a one-class SVM baseline. The same paper concedes the cost of its own geometry: the Poincaré embedding introduces significant computational overhead on large graphs, and the model is the slowest in its own comparison table, at roughly 34 to 39 seconds per thousand records. A purported scaling lever here turns into a throughput penalty, which is a useful caution for the whole branch.

Classical learning supplies the branch’s counterpoint. One paper scales least-squares regression using fast spectral embedding and random Fourier feature mapping [4]. Random Fourier features approximate a shift-invariant kernel so that a kernel least-squares problem becomes linear in a finite-dimensional feature space, and the spectral embedding handles the manifold structure. This is the corpus’s only non-deep estimator, and it is the only item whose method is a classical statistical one. Its abstract is not indexed, so the review can describe the approach from the title but cannot state its datasets, its complexity, or its results. That limit is recorded rather than papered over, and it is one reason Section 2.6 treats evidence availability as a finding in its own right.

Interpretation cost is the branch’s last resource. One paper builds PEXP, a parallel tree-based post-hoc explainer for models trained on large-scale data [5]. The design generates a localized sample set around a target instance through data-distribution-aware perturbation, weights the samples with a kernel function, and builds parallel ensemble trees as an interpretable surrogate using ideas from bagging and boosting. Feature-importance explanations are aggregated across samples to give a global view. The paper’s pitch is that computational inefficiency is the real obstacle to explaining large models, which makes explanation cost a scalability problem rather than a presentation problem. As with the classical estimator, the abstract gives no dataset, no platform, no metric, and no named baseline, so the claimed advantages in runtime and explanation quality are unquantified here. The paper’s single case study is video anomaly detection in a smart-city setting, which overlaps the streaming-detection territory of the space-tree work [2]; the pairing makes the branch’s split clear, since one paper explains a detector while the other builds one that needs no retraining.

The branch closes on a method arc rather than a single result. A review of multimodal data integration in cancer research traces the field from classical learners through deep latent fusion to foundation models, and it reports that intermediate and attention-based latent fusions outperform early concatenation and late voting in

survival and metastasis prediction [6]. The same review documents a severe translational gap, with more than 90% of the work it surveys resting on retrospective cohorts and near-zero prospective trials. Its role here is to supply the trajectory that the branch’s individual papers occupy: purpose-built transformers [1] and adaptive streaming ensembles [2] are the deep stage, while the corpus’s language-model items in the applications branch are the emerging stage. Fig. 4 shows that shift across the corpus by journal.

2.4 Platforms, pipelines, and data engineering

Branch B2 holds the machinery under the models, and it embodies “big data” in the strictest sense of the corpus. Its three papers cover the economics of distributed training [7], a cluster engine [8], and a streaming tier [9]. Each names a real platform layer or a real cost, and together they outline an infrastructure stack. No single application paper in the corpus deploys that stack end to end, which is itself a structural observation about the field’s division of labour.

The economics layer prices federated training. One paper analyses the cost of federated learning for 6G networks, measuring training time, communication overhead, and energy consumption across federated approaches [7]. Federated learning keeps data on user devices, so the paper reframes distributed training as an economics and energy question. Its reported conclusion is that federated learning can accelerate training and cut data transferred across the network, but that the benefit depends on the specific approach and the network conditions. No timing, energy, or bandwidth figures, and no named datasets or algorithms, appear in the indexed abstract, so the claim stands as a direction rather than a quantified result. The paper’s value for this review is the vocabulary it supplies. It is the only item that prices federated training, and it therefore sets the terms for the privacy–scale tension that Section 2.6 develops.

The cluster layer appears in one paper that detects opinion fraud on massive datasets using Apache Spark [8]. Spark is the only computing platform named, and it comes from the title. This item has no indexed abstract in any source that was checked, and the publisher page returns a script shell rather than text. As a result the review can record the cluster setting and the fraud-detection target but cannot describe the algorithm, the dataset, or any result. The paper is the corpus’s single unambiguous cluster-computing contribution, and it is the highest-priority retrieval gap in the set. Its presence at title level still matters: it shows that classic cluster stacks continue to anchor production workloads even as the corpus turns toward streaming and language models.

The streaming layer is more fully documented. One paper builds a cloud-based system for real-time prediction of systolic blood pressure and heart rate [9]. The pipeline is explicit: simulated sensors publish to an Apache Kafka topic, Spark Streaming processes sliding windows at a fog layer, and a cloud layer stores history and retrains models. The forecasting model is a temporal convolutional network, chosen over LSTM, GRU, and sequence-to-sequence alternatives, and the multi-task variant predicts both vital signs jointly. The headline results for the eight-minute horizon are a heart-rate RMSE of 1.5428 and MAE of 1.0871, and a systolic-pressure RMSE of

4.1446 and MAE of 2.4323, with the temporal convolutional network best at all horizons and significantly better than the recurrent baselines. Multi-task learning beats single-task learning across horizons. Two caveats bound these numbers. The evaluation uses minute-by-minute data from only two patients, so the significance tests are within-patient rather than cross-patient. The online pipeline is tested on simulated sensors, which means the latency figures are architectural rather than measured clinical latency. The paper also defers privacy explicitly, naming federated learning as future work.

Two patterns close the branch. First, the three layers do not connect: the cost model, the cluster engine, and the streaming tier appear in separate papers, and no paper integrates them. Second, real-time claims are almost always evaluated on architecture rather than on measured operational latency. The streaming paper’s sensors are simulated [9], the coverage of streaming in the methods branch comes from abstract-level thresholds [2], and the cluster paper cannot be read at all [8]. The corpus is building streaming systems faster than it is measuring them.

2.5 Applications and domains

Branch B3 is the corpus’s largest, with eight papers spread across security, telecom, finance, agriculture, affective computing, and industrial monitoring. The domains are far apart, but a pattern holds across them. Where the data has a native graph or stream structure, most of the method-building happens. Where it does not, applications mostly consume off-the-shelf deep learning. The branch is also where the “big data” label loosens most visibly, because application papers adopt the term for data variety and image volume rather than for distributed scale.

Security supplies the first vertical. One paper detects malware from program representations, using control-flow and function-call graphs, and it attacks two problems at once [10]. The first is graph size: the representations are large and complex, so the paper applies novel graph-reduction techniques before learning. The second is explanation: a binary classification is not enough for an analyst, so the framework applies GNNExplainer to surface important subgraphs. The authors report that the reduction cuts graph size and complexity while maintaining detection performance, and that the extracted subgraphs give analysts better insight into the model’s decision. The abstract reports no numbers, no dataset, and no baseline, so the efficiency claim is unquantified here. The paper’s place in the corpus is clear regardless: it is the only item where the data structure itself is the obstacle and where explanation is an analyst-facing artifact rather than a feature ranking. That makes it the application-bound partner of the parallel explainer in the methods branch [5].

The same vertical has a survey counterpart. A PRISMA review of anomaly detection for telecom networks charts the move from rule-based systems to deep learning [11]. Its framing is blunt: rule-based detection is no longer effective in a fast-evolving telecom landscape. It catalogues classical methods, deep methods, and emerging techniques, and it reports that deep models reach roughly 85 to 97% detection quality while paying in latency, with autoencoders in the range of two thousand to seven thousand milliseconds against 100 to 400 milliseconds for isolation forests. Its central operational claim is that deployment, not accuracy, is the blocker: compute cost, edge

latency, model drift, and retraining burden dominate. The review runs no experiments of its own and states that no datasets were generated or analysed. Its accuracy and latency ranges are aggregate and unattributed to named datasets, so they are read here as indicative. It is also the corpus’s only treatment of operator-facing economics, and it identifies a language-model-sized gap: the survey has no foundation-model content at all, which the later language-model items in this corpus fill.

Finance supplies the finance-domain case, and it is the corpus’s cleanest example of graph learning on natively relational data. One paper performs risk assessment for internet financial companies using company big data and a heterogeneous graph [12]. The graph is the natural schema of the business records rather than an engineered artifact, which distinguishes it from program graphs [10] and network-traffic graphs [3]. The abstract is not indexed and the green preprint was blocked at retrieval, so the review cannot describe the model, the data, or the results. The paper is included here for its structural position: it extends graph-based learning from security and network domains into finance, supporting the pattern that native relational structure attracts method-building.

Agriculture shows the label loosening. One paper detects plant disease at scale through a deep-learning framework that it presents as big-data analytics [13]. The model is a NASNetLarge backbone with the first 80 layers frozen and a custom classification head, trained with AdamW and mixed precision, and it classifies 12 severity classes across two crops. It reaches 97.33% accuracy on a merged thirty-thousand-image dataset after 26 epochs, beating DenseNet201 at 96.40%, InceptionResNetV2 at 95.70%, and ResNet152V2 at 93.23%, and it reaches 95.60% on the wheat dataset alone. A Vision Transformer baseline reaches only 41.3%, which the authors attribute to the model’s data hunger. The infrastructure tells a different story from the title. The work runs on a single Kaggle GPU with no cluster engine and no streaming platform, and the dataset is image volume rather than distributed scale. The paper also reports an ablation showing that removing its training enhancements drops accuracy to 82.36%, which is the corpus’s clearest evidence that a training recipe, not just an architecture, drives results. Real-field validation is left undone, which the authors acknowledge.

Affective computing supplies the branch’s most elaborate method and its most conspicuous reporting problem. One paper performs multimodal emotion recognition through prompt engineering and deep adaptive learning [14]. It fuses text, audio, video, and motion through a ten-phase pipeline that uses a Mistral-7B text encoder, HuBERT for audio, a LLaVA and TimeSformer pair for video, and a pose model for motion, then combines them through adaptive graph networks, a cross-modality transformer, and prototypical contrastive learning. Inference is exported through an ONNX runtime, and the reported inference cost is 0.3 to 0.5 milliseconds per sample. The reported accuracies are 99.82% on IEMOCAP and 99.81% on MELD, with cross-lingual results of 99.1 to 99.4% on a multilingual benchmark that was never used for training. Those numbers sit far above the published range for emotion recognition, where competitive systems cluster near 70%; the paper’s own comparison table cites competitors at that level. The gap points to a likely evaluation problem, such as label leakage or a non-speaker-independent split, rather than a capability leap. This review

therefore treats the figures as reporting-practice evidence and does not quote them as state of the art. Fig. 3 plots them at face value but flags the anomaly. The paper also concedes that attention mechanisms give only partial interpretability and that dedicated post-hoc explanation is absent.

Foundation-model and agentic applications close the branch, and they expose a grounding gap. One paper builds TS-MLLM, a multimodal large language model for industrial time-series analysis [15]. It combines patch modelling of temporal signals, a spectrum-aware vision-language adaptation for frequency-domain images, and a temporal-centric attention fusion that retrieves visual and textual cues using temporal features as queries. The authors claim to be first to model temporal signals, frequency-domain images, and textual knowledge jointly, and they report that the framework outperforms state-of-the-art methods, especially in few-shot and complex scenarios. The full text is paywalled and no baseline, dataset, or metric is named in the abstract, so the review describes the design without endorsing the magnitude. A companion paper couples graph learning with a language model to improve graph reasoning [16]. It replaces the usual conversion of graphs into text with an end-to-end coupling through a dynamic task configuration system, using local structure analyzers and global pattern synthesizers. It reports an average accuracy improvement of 54.44% across four graph-reasoning tasks and a context reduction of 96.45% against a text-serialization baseline. The context reduction is the corpus’s only quantification of prompt-window pressure, and it is directly relevant to the scaling problem; the accuracy figure is an aggregate over unnamed tasks, so it is reported here as a summary rather than a per-task result. Both papers inherit the same limit: neither tests its system under the streaming, cluster, or memory constraints that Section 2.4 treats as first-class.

A survey of agentic AI supplies the branch’s analytical hinge and the corpus’s clearest statement of the domain divide [17]. It sorts the field into symbolic architectures, which rely on planners and persistent state, and neural architectures, which rely on stochastic generation and prompt-driven orchestration, and it argues that conflating the two obscures how modern agents work. Its finding is that paradigm choice is domain-driven: symbolic and hybrid designs dominate safety-critical work such as healthcare and robotics, while neural systems dominate data-rich adaptive fields such as finance and education. The survey also warns that neural agents depend on expensive cloud compute, and that context-window limits produce agents with no memory across sessions. The warning matters here because the corpus’s agentic and foundation-model work shows no big-data grounding. None of those items evaluates under the platform constraints that the infrastructure branch treats as central. The scale vocabulary shifts from data to model size, and the operational questions that the rest of the corpus asks go unasked.

Security and anomaly detection is the single most-represented domain across the corpus. Malware detection [10], telecom anomaly detection [11], network-traffic detection from the methods branch [3], streaming fraud detection [2], and the Spark fraud paper [8] all sit in that vertical, whether their methods are graph-based, streaming, or platform-centric. That concentration suggests where the field’s shared problems are

hardest, and it also explains why metrics differ so widely across the branch: a malware graph, a network flow, and a credit-card transaction do not admit one common benchmark.

2.6 Cross-cutting concerns and evaluation practice

Three cross-cutting patterns cut across the branches, and they carry the review’s analytical weight. The first is a tension between pooling data and protecting it. The second is fragmented and often irreproducible evaluation. The third is explainability as a recurring requirement that the corpus has not solved.

The privacy–scale tension is the corpus’s defining contradiction. One program of work assumes that capability comes from centralizing and pretraining on everything available, which is the premise behind the foundation-model items [15, 16] and the multimodal fusion work [6]. Another program assumes that data cannot move at all. Federated learning keeps training on the device, and the corpus prices the cost: the federated cost analysis measures training time, communication overhead, and energy consumption for 6G [7], and it finds the benefit conditional on the approach and the network. Privacy-preserving federated learning does not come free. An adaptive aggregation framework for healthcare reports up to 96.3% accuracy while cutting execution time by about 20% and rounds by about 20% [18]. It switches between averaging and gradient-based updates according to a divergence statistic, and it reports a full privacy–utility trade-off: at a noise scale giving the strongest privacy, accuracy falls from 96.5% to 94.5%. The framework is the corpus’s only source of a privacy–utility table for federated healthcare learning, and it is also the only item that reports a cost for locality. Its own limits are material. All federation is simulated on one machine, differential privacy is applied only to the centralized arm rather than to the federated rounds, and the datasets total roughly fourteen thousand images, which is institutional heterogeneity rather than volume. The tension has no resolution in the corpus. The streaming healthcare system defers privacy to future work entirely [9], choosing latency over locality, so the two healthcare papers pull in opposite directions and neither reconciles them.

Synthetic data is the second privacy mechanism, and its evaluation problem is general. A review of generative models for healthcare synthetic data compares five families across imaging, structured records, text, signals, and genomics [19]. It finds that no family wins everywhere: diffusion models lead on image fidelity, generative adversarial networks give measured gains when used for augmentation, and variational autoencoders handle structured records best. Its most transferable finding is that fidelity metrics do not track usefulness. Visual scores such as FID and SSIM, the review reports, do not always align with downstream diagnostic performance or clinical trustworthiness. That conclusion joins a broader problem. The same review reports that gradients, outputs, and auxiliary data each create a leakage surface, and that regulators treat synthetic data as supplementary evidence rather than sufficient grounds for approval. Its own method is a structured survey rather than a systematic review, with no exclusion counts and no numerical quality scores, so its comparative table mixes peer-reviewed results, preprints, and abstract-only claims. The review is

candid about that, and the candour is itself part of the evidence: even the governance literature reports its own evidence standard unevenly.

Evaluation fragmentation is the cross-cutting pattern with the strongest empirical support. The corpus contains exactly one reproducibility audit, and it is decisive for its scope. A survey of deep multivariate time-series models re-runs eight official codebases under one protocol, using a fixed seed baseline and then multiple seed runs [20]. Broad trends reproduce; exact numbers often do not. One language-model-based forecaster falls from an F1 of 82.26 to 79.25 on one dataset while holding steady on another, and its classification results drop by several points on three gesture and heartbeat datasets. A cross-modal model fails to reproduce in all 16 of its dataset-and-horizon settings. A state-space model drops from 87.00% to 42.82% on one classification dataset. A graph model’s RMSE rises from 11.62 to 12.39, just past the audit’s two-sigma threshold. The audit’s summary judgement is that single-run evaluations often overestimate model stability and that routine variance reporting is needed. The audit is deliberately limited: eight models, one protocol, one GPU, and a descriptive threshold rule rather than a formal significance test. Those limits matter for how far the claim travels. Even bounded that way, the finding reaches the rest of the corpus. Headline figures elsewhere are reported as single runs [1], as threshold statements from abstracts [2], or as unattributed ranges [11], and the audit shows why such reporting is fragile. It is also the corpus’s clearest case of accountability lagging capability: the most-cited item in the set is a survey with 134 citations, while the only audit has none [20].

Explainability is the third pattern, and it remains largely unsolved. The requirement appears in every branch, which is itself telling. A parallel explainer attacks the cost of explaining large models [5], a malware framework treats analyst-facing sub-graphs as part of the pipeline [10], the affective-computing paper concedes the absence of post-hoc explanation [14], the streaming clinical system lists interpretability as future work [9], and the synthetic-data review links opacity to clinical trust [19]. Two of those items make explanation a first-class design goal rather than a diagnostic add-on, and they answer different questions: one explains a model’s surrogate behaviour at scale [5], and the other explains a data structure to a human analyst [10]. The corpus has no shared measure for explanation quality. Fidelity is asserted rather than measured where a number would be needed, and no paper reports an explanation-faithfulness benchmark. The pattern that emerges is not that explainability is ignored; it is that explainability is required everywhere and standardized nowhere.

The corpus’s structural limits belong in this section as findings. No selected paper addresses reinforcement learning at scale, and no selected paper makes a purely theoretical contribution. Part of the 2025–2026 record is also inaccessible: ten of the 20 papers could not be read in full, spanning abstract-only items [1, 2, 5, 7, 10, 15, 16] and items with no indexed abstract at all [4, 8, 12]. That inaccessibility is uneven across branches, and it constrains what any review of this corpus can claim. Scalability, finally, is frequently architectural rather than measured. The parallel explainer names no platform [5], the federated cost analysis names no figures [7], and the agriculture paper’s “large-scale” dataset runs on one GPU [13]. Claims of that shape are not false, but they are unverified, and the audit shows what verification costs.

2.7 Future directions

The directions below are drawn from the observations in Section 2.6 rather than listed generically. Each one names the gap it addresses.

Reinforcement learning at scale is absent from the taxonomy, and that absence is a method cell waiting to be filled. The corpus covers supervised, unsupervised, and generative learning at various scales, but no item learns a policy over a large environment with the platform and evaluation discipline the rest of the corpus demands. Whether that reflects the field or reflects this corpus is an open question, and the answer requires a broader search than the one performed here.

Agentic systems need big-data grounding. The agentic survey warns that neural agents depend on unsustainable cloud compute and that context limits break continuity across sessions [17]. The corpus’s language-model items show the same pattern: capable systems evaluated on benchmarks that do not test streaming, memory, or cluster constraints [15, 16]. Building agents that operate over distributed data, rather than over text, is the direction those two observations imply.

The privacy–scale contradiction is unpriced beyond one cost analysis. The federated cost work measures time, communication, and energy [7], and the adaptive aggregation framework prices the accuracy cost of differential privacy [18], but no paper puts both sides on one scale. A study that reports capability, privacy budget, and platform cost together would settle a question the corpus currently leaves to inference.

Accountability lags mechanism. The corpus holds two privacy-oriented works and one synthetic-data review against a single reproducibility audit [18–20], and that imbalance means the field’s mechanisms are more advanced than its checks. Routine variance reporting, per-seed results, and disclosed preprocessing would raise the evidence standard without new methods, and the audit shows the reporting template is already known [20].

Explanation at scale is single-sourced. Only one paper treats explanation cost as the scalability bottleneck [5], and only one other fuses explanation into an application pipeline [10]. Explanation-faithfulness benchmarks and shared metrics are missing, so the value of an explanation cannot be compared across systems. That gap sits directly under the corpus’s most frequently stated requirement.

3 Tables and Figures

This section presents the comparison tables and the five figures that summarize the corpus, and it explains what each item shows. Table 1 is the master comparison of all 20 papers. It lists each paper by citation number, authors and year, journal, method family, the data and scale it reports, and the headline result or an explicit note that the result is not available. The “not available” entries are deliberate. Six papers have no comparable metric that this review can verify, and several more are described only at the level their abstract supports. Table 2 maps the five taxonomy branches to their papers and gives a representative contribution for each branch.

Fig. 1 plots the corpus landscape. It places all 20 papers on two curated axes, one for data scale and kind, and one for the binding constraint, with colour marking the journal and marker shape distinguishing research articles from review articles.

The journal and article-type attributes are exact. The axis positions are interpretive, documented point by point in the figure’s data-source file, and they are a framing device rather than a measurement. The figure makes the review’s organizing claim visible: the corpus splits into a lower band that learns on more data and an upper band that learns under constraints.

Fig. 2 is the taxonomy. It shows the two axes and the five branches B1–B5, with every paper placed exactly once as a citation-number chip. The placement totals 20, and the figure marks the empty cells that Section 2.6 discusses. Fig. 4 shows the method-generation shift across the corpus by journal, with “not specified” kept visible rather than guessed; one Big Data Research item is unlabeled because its abstract is not indexed [8], and the small counts per journal should be read as counts rather than proportions.

Fig. 3 puts the corpus’s reported results side by side, and it must be read with care. Values come from different datasets and different metrics, so they are reported anchors rather than a like-for-like ranking. Papers with no comparable metric appear in an explicit text band rather than at zero, and several reviews are marked as qualitative claims only. Threshold values from abstracts, such as those from the streaming-detection work [2], are plotted at the threshold and hatched to mark them as non-precise. One point deserves emphasis: the 99.8% accuracies from the multi-modal emotion work [14] are plotted at face value but are anomalously high against the emotion-recognition literature, and they are flagged here as a reporting concern rather than a capability result. Panel (b) shows regression-error anchors and the corpus’s only reported-to-reproduced pair, where the reproducibility audit raises a published RMSE from 11.62 to 12.39 for the same model and dataset [20].

Fig. 5 returns to the selection method from the Introduction. It shows the funnel from 1,519 retrieved items through the keyword screen at 441 and Crossref verification at 49 to the 20 included papers, with the per-journal split. The figure is descriptive and carries no performance claim. The axis is log-scaled and labelled as such.

Two consistency rules apply across these items. Every figure and table is cited from the running text, and the citation numbers in Table 1 match the paper’s own numbering, not the internal identifiers used during screening. The figures themselves never display internal identifiers.

4 Conclusion

This review examined 20 papers from four journals, five from each, and synthesized them thematically. The corpus splits into five branches: scalable learning methods, platforms and pipelines, applications and domains, foundation models and language models, and privacy, governance, and evaluation. Across those branches, one claim held up. Big-data machine learning in this corpus has moved from learning on more data to learning under constraints, and those constraints are privacy, platform, and proof.

Three takeaways follow. The first is that constraint beats volume as the organizing problem. Four of the five methods papers optimize a resource rather than an accuracy leaderboard: computation for the explainer and the classical estimator, latency for

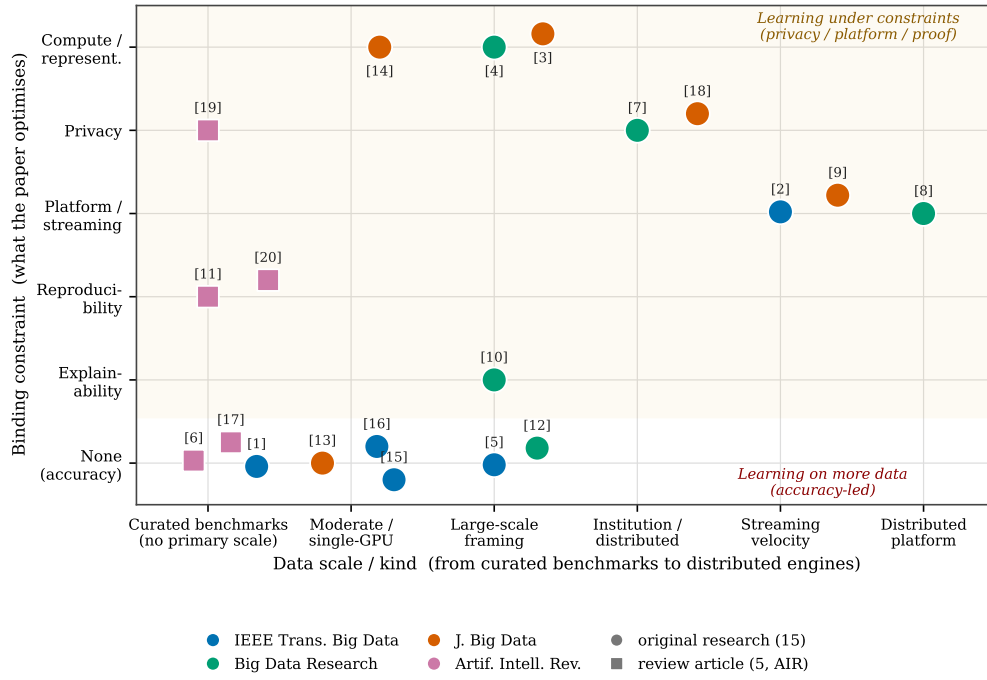


Fig. 1 Corpus landscape. All 20 papers are placed on two curated axes: data scale and kind (x) and binding constraint (y). Colour marks the journal and marker shape distinguishes research articles from review articles. Journal and article type are exact; the axis positions are interpretive design coordinates documented in the figure data-source file. The corpus splits into a lower band that learns on more data and an upper band that learns under constraints.

the streaming detectors, and representation efficiency for the graph embedding. The applications papers show the other side of the same coin, since several of them use “big data” for data variety or image volume while running on a single machine. The second takeaway is that privacy and scale collide, and the corpus does not resolve the collision. Federated learning cuts data movement and protects locality, and it prices the cost: about 20% savings in time and rounds in one framework, and an accuracy drop of two points under strong differential privacy. Centralized training assumes the opposite, and the streaming healthcare system defers privacy entirely. The third takeaway is that evaluation is the weakest joint. The corpus’s only reproducibility audit shows that broad trends survive re-runs while exact numbers often do not, and the rest of the corpus reports single runs, abstract-level thresholds, or unattributed ranges.

The review has limits, and they bound the claim. The corpus is 20 papers from four journals in one two-year window, all in English, and the theme mix is curated rather than sampled, so the branch proportions are not the field’s proportions. Ten of the 20 papers could not be read in full, three of them have no indexed abstract at all, which means several claims are reported at the level their available evidence supports rather than at full strength. Absence claims, such as the missing reinforcement-learning work, are scoped to this corpus and not to the field. A future review that reads the

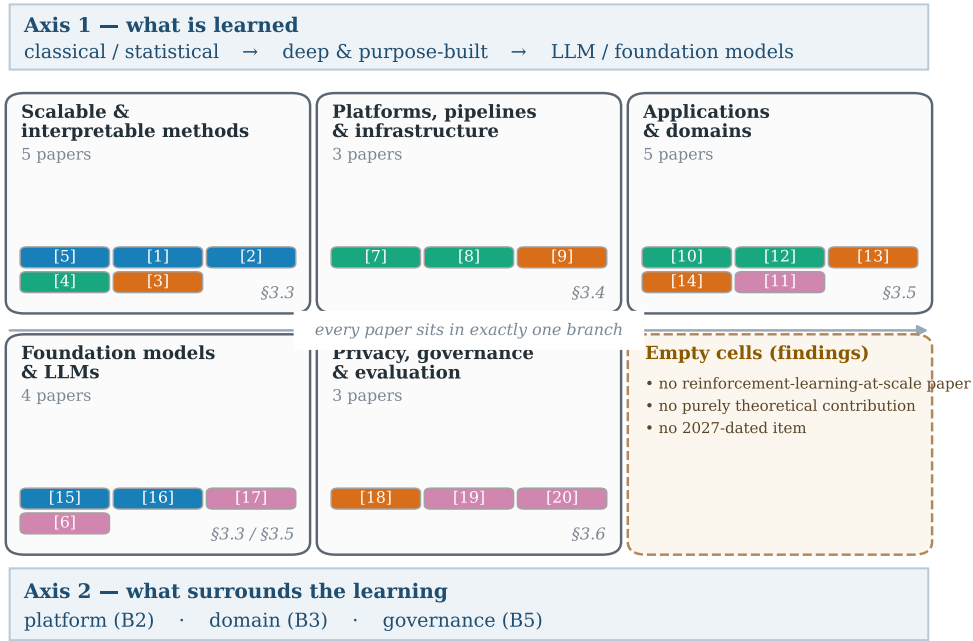


Fig. 2 Two-axis taxonomy of the corpus. Axis 1 is what is learned; Axis 2 is what surrounds the learning. Five branches B1–B5 place each of the 20 papers exactly once, shown as citation numbers, for a total of 5+3+5+4+3. The branch assignment is this review’s classification of this curated corpus, and the empty cells (no reinforcement learning at scale, no purely theoretical contribution, no 2027 item) are findings rather than search gaps.

inaccessible items or widens the venue set could confirm or revise what this one reports. What this review can state is narrower and firmer: within a curated 2025–2026 corpus of four journals, the field’s binding limits are privacy, platform, and proof, and the corpus’s own evidence, not the popular framing, shows it.

Acknowledgements. Not applicable.

Declarations

- Funding: Not applicable.
- Conflict of interest: The authors declare no competing interests.
- Ethics approval and consent to participate: Not applicable.
- Consent for publication: Not applicable.
- Data availability: The corpus is bibliographic. The 20 reviewed papers are identified by DOI in the reference list. Figure source data is documented in the project’s figure data-source file.
- Materials availability: Not applicable.
- Code availability: The figures were generated by a reproducible script. No other code was developed for this review.

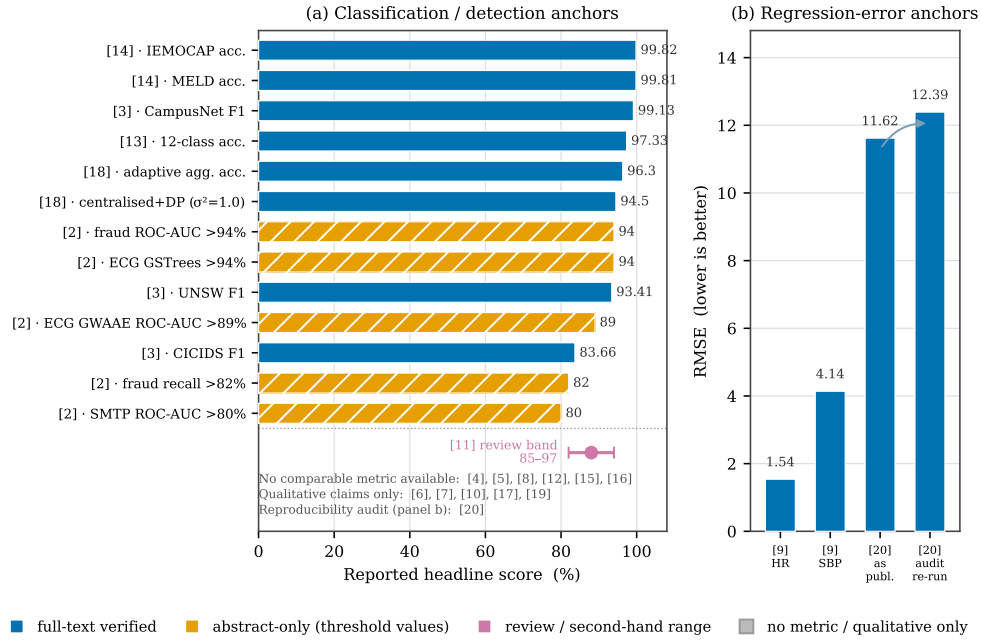


Fig. 3 Reported-result anchors, not a like-for-like ranking. Panel (a) shows classification and detection values and panel (b) shows regression error with the corpus’s only reported-to-reproduced pair. Values come from different datasets and metrics, so they are anchors rather than a leaderboard. Papers with no comparable metric are listed in an explicit text band and are never plotted as zero. Hatching marks threshold values quoted from abstracts. The [14] accuracies (99.82% and 99.81%) are plotted at face value but are anomalously high against the emotion-recognition literature and are flagged as a reporting concern, not a capability result.

- **Author contribution:** All authors contributed to the study. The corresponding author coordinated the review. Conception and design, literature screening, evidence extraction, thematic synthesis, and manuscript drafting were divided across the author team as recorded in the project tracker. All authors read and approved the final manuscript.

References

- [1] Xiao Z, Xing H, Qu R, et al (2025) Knowledge aggregation transformer network for multivariate time series classification. *IEEE Transactions on Big Data* 11(6):3413–3429. <https://doi.org/10.1109/tbdata.2025.3594294>, URL <https://doi.org/10.1109/tbdata.2025.3594294>
- [2] Gunbilek E, Akusta Dağdeviren Z, Bulut H (2026) Enhanced approaches for anomaly detection in streaming data: Coupling Gaussian distributions with space trees and adaptive AutoEncoders. *IEEE Transactions on Big Data* 12(4):1318–1331. <https://doi.org/10.1109/tbdata.2026.3679567>, URL <https://doi.org/10.1109/tbdata.2026.3679567>

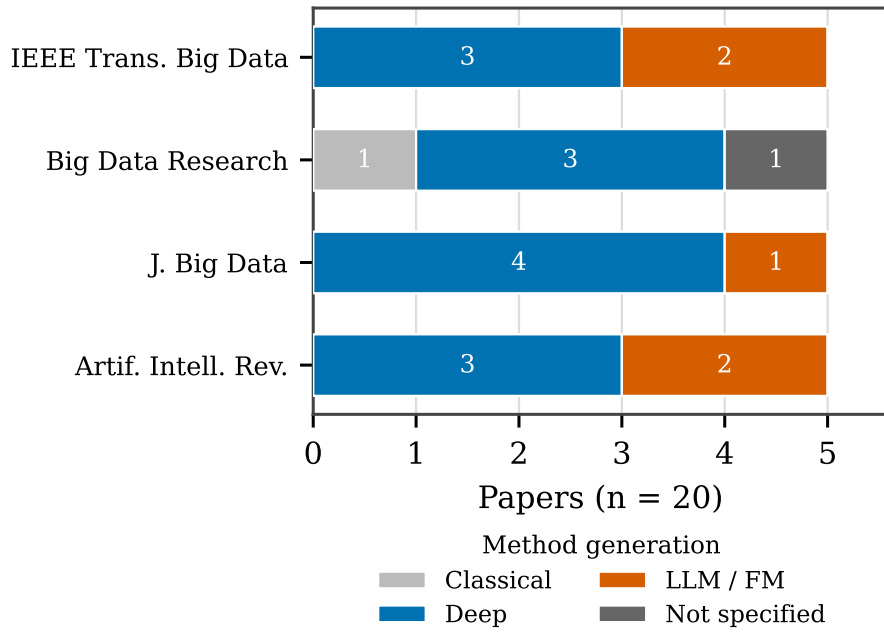


Fig. 4 Method-generation shift across the corpus by journal. Bars count the papers in each journal by method generation. The label is a curated classification per paper, and “not specified” is kept visible rather than guessed, because one Big Data Research item has no indexed abstract. Each journal contributes five papers, so the bars are read as counts, not proportions.

doi.org/10.1109/tbdata.2026.3679567

- [3] Zhang H, Zhou Y, Xu H, et al (2025) Graph neural network approach with spatial structure to anomaly detection of network data. Journal of Big Data 12(1):105. <https://doi.org/10.1186/s40537-025-01149-y>, URL <https://doi.org/10.1186/s40537-025-01149-y>
- [4] Li X, Liu J (2026) Large-scale least squares regression based on fast spectral embedding and random Fourier feature mapping. Big Data Research 43:100589. <https://doi.org/10.1016/j.bdr.2026.100589>, URL <https://doi.org/10.1016/j.bdr.2026.100589>
- [5] Jiang W, Chang CY, Huang TC, et al (2026) PEXP: A scalable parallel tree-based framework for interpreting models on big data. IEEE Transactions on Big Data 12(4):1177–1194. <https://doi.org/10.1109/tbdata.2026.3668673>, URL <https://doi.org/10.1109/tbdata.2026.3668673>
- [6] Muneer A, Waqas M, Saad MB, et al (2026) From classical machine learning to emerging foundation models: review on multimodal data integration for cancer research. Artificial Intelligence Review 59(4). <https://doi.org/10.1007/s10462-026-11522-9>, URL <https://doi.org/10.1007/s10462-026-11522-9>

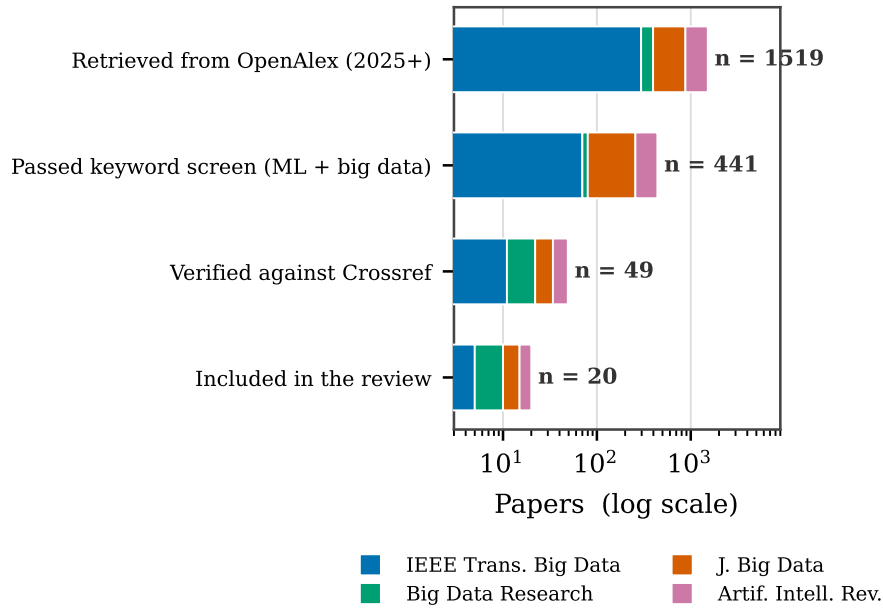


Fig. 5 Selection flow for the corpus, in the style of a PRISMA funnel. Counts run from 1,519 items retrieved from OpenAlex (2025 or later) through a keyword screen at 441 and Crossref verification at 49 to 20 included papers, with a per-journal split of 5/5/5/5. The x-axis is log-scaled and labelled as such. The figure is descriptive and carries no performance claim.

- [7] Teixeira R, Almeida L, Antunes M, et al (2025) Efficient training: Federated learning cost analysis. Big Data Research 40:100510. <https://doi.org/10.1016/j.bdr.2025.100510>, URL <https://doi.org/10.1016/j.bdr.2025.100510>
- [8] Ghodsi S, Moeini A (2026) Opinion fraud detection on massive datasets by spark. Big Data Research 43:100590. <https://doi.org/10.1016/j.bdr.2026.100590>, URL <https://doi.org/10.1016/j.bdr.2026.100590>
- [9] Saleh H, El-Rashidy N, Mostafa S, et al (2025) Cloud based real-time multivariate multi-step prediction of systolic blood pressure and heart rate using temporal convolutional network and Apache Spark. Journal of Big Data 12(1):170. <https://doi.org/10.1186/s40537-025-01207-5>, URL <https://doi.org/10.1186/s40537-025-01207-5>
- [10] Mohammadian H, Higgins G, Ansong S, et al (2025) Explainable malware detection through integrated graph reduction and learning techniques. Big Data Research 41:100555. <https://doi.org/10.1016/j.bdr.2025.100555>, URL <https://doi.org/10.1016/j.bdr.2025.100555>
- [11] Edozie E, Shuaibu AN, Sadiq BO, et al (2025) Artificial intelligence advances in anomaly detection for telecom networks. Artificial Intelligence Review 58(4). <https://doi.org/10.1007/s10462-025-11108-x>, URL <https://doi.org/10.1007/s10462-025-11108-x>

- [12] Liu Q, Qiu Y, Jia Y, et al (2026) Heterogeneous graph-based risk assessment for internet financial companies with company big data. *Big Data Research* 45:100620. <https://doi.org/10.1016/j.bdr.2026.100620>, URL <https://doi.org/10.1016/j.bdr.2026.100620>
- [13] Elfouly MK, Abdelaziz A, Gomaa WH, et al (2025) A deep learning-based framework for large-scale plant disease detection using big data analytics in precision agriculture. *Journal of Big Data* 12(1):205. <https://doi.org/10.1186/s40537-025-01265-9>, URL <https://doi.org/10.1186/s40537-025-01265-9>
- [14] Wafa AA, Eldefrawi MM, Salah Farhan M (2025) Advancing multimodal emotion recognition in big data through prompt engineering and deep adaptive learning. *Journal of Big Data* 12(1):210. <https://doi.org/10.1186/s40537-025-01264-w>, URL <https://doi.org/10.1186/s40537-025-01264-w>
- [15] Wang H, Li Y, Zhu YC, et al (2026) TS-MLLM: A multi-modal large language model-based framework for industrial time-series big data analysis. *IEEE Transactions on Big Data* <https://doi.org/10.1109/tbdata.2026.3695338>, URL <https://doi.org/10.1109/tbdata.2026.3695338>, online first (early access); volume and pages not yet assigned
- [16] Chai Z, Zhang T, Wu L, et al (2025) GraphLLM: Boosting graph reasoning ability of large language model. *IEEE Transactions on Big Data* 12(2):475–483. <https://doi.org/10.1109/tbdata.2025.3627488>, URL <https://doi.org/10.1109/tbdata.2025.3627488>
- [17] Abou Ali M, Dornaika F, Charafeddine J (2025) Agentic AI: a comprehensive survey of architectures, applications, and future directions. *Artificial Intelligence Review* 59(1). <https://doi.org/10.1007/s10462-025-11422-4>, URL <https://doi.org/10.1007/s10462-025-11422-4>
- [18] Haripriya R, Khare N, Pandey M, et al (2025) A privacy-enhanced framework for collaborative big data analysis in healthcare using adaptive federated learning aggregation. *Journal of Big Data* 12(1):113. <https://doi.org/10.1186/s40537-025-01169-8>, URL <https://doi.org/10.1186/s40537-025-01169-8>
- [19] Waseem HM, ul Islam S, Matragkas N, et al (2025) Review of generative AI for synthetic data generation: a healthcare perspective. *Artificial Intelligence Review* 59(2):55. <https://doi.org/10.1007/s10462-025-11440-2>, URL <https://doi.org/10.1007/s10462-025-11440-2>
- [20] Vats V, Kim H, Linares E, et al (2026) A survey of deep multivariate time-series models with an empirical reproducibility audit. *Artificial Intelligence Review* <https://doi.org/10.1007/s10462-026-11674-8>, URL <https://doi.org/10.1007/s10462-026-11674-8>, online first; volume and pages not yet assigned

Table 1 Corpus summary: the 20 reviewed papers in citation order. “Not available” marks an item whose full text could not be read or whose abstract is not indexed; no unverified value is entered.

Ref.	Authors, year	Journal	Method family	Data and scale	Headline result
[1]	Xiao et al., 2025	IEEE TBD	Aggregation transformer (local + global)	13 UEA archive datasets; rank-based protocol	Lowest avg. rank vs 6 SOTA transformers (W/T/L 9/6/15); no per-dataset accuracies or variance
[2]	Gunbilek et al., 2026	IEEE TBD	Gaussian space trees + adaptive-window autoencoder; streaming	ECG5000, credit-card fraud, SMTP streams; sizes not stated	Threshold values only from abstract: >94% (ECG), recall >82% and ROC-AUC >94% (fraud); abstract-level
[3]	Zhang et al., 2025	JoBD	Hyperbolic (Poincaré-ball) graph embedding + GAT; unsupervised	CICIDS2017 3.12M records; UNSW-NB15 2.54M; campus log 1.25M	F1 0.8366 / 0.9341 / 0.9913; edge weights $\approx 2\times$ node augmentation; slowest model in its own tables
[4]	Li and Liu, 2026	BDR	Fast spectral embedding + random Fourier features; classical regression	Not available; abstract not indexed	Not available; abstract not indexed
[5]	Jiang et al., 2026	IEEE TBD	Parallel tree-based post-hoc explainer (bagging/boosting surrogate)	Not available; no dataset or platform named	Qualitative claims only; no metrics or baselines in abstract; abstract-level
[6]	Muneer et al., 2026	AIR	Review: classical to deep to foundation-model multimodal fusion	Secondary: PRISMA review of 54 studies; cohorts to 110M cells	Intermediate and attention fusion beat early/late; >90% retrospective, near-zero prospective trials
[7]	Teixeira et al., 2025	BDR	Federated learning cost analysis (time, communication, energy)	Not available; no dataset or device count in abstract	Qualitative claims only; FL accelerates training and cuts transfer, contingent on approach and network; abstract-level
[8]	Ghodsai and Moeini, 2026	BDR	Apache Spark cluster pipeline (title-level only)	Not available; abstract not indexed	Not available; abstract not indexed
[9]	Saleh et al., 2025	JoBD	Temporal convolutional network + Kafka and Spark Streaming; fog/cloud	MIMIC-III, minute-level SBP and HR for 2 patients	Multi-task TCN 8-min: HR RMSE 1.5428, SBP RMSE 4.1446; beats LSTM/GRU at $p < 0.05$
[10]	Mohammadian et al., 2025	BDR	Graph reduction + GNN detection + GNNExplainer subgraphs	Not available; no dataset or graph scale in abstract	Qualitative claims only; reduction maintains detection performance; abstract-level
[11]	Edozie et al., 2025	AIR	Review: rule-based to deep learning telecom anomaly detection	No own data; PRISMA screen of 3,084 to 160 articles	Deep models 85–97% with latency cost; deployment, not accuracy, is the blocker; ranges unattributed
[12]	Li et al., 2026	BDR	Hybrid	Not available	Not available

Table 2 Taxonomy mapping: branch, papers, and a representative contribution. Each paper appears in exactly one branch.

Branch	Theme	Papers	Representative contribution
B1	Scalable and interpretable learning methods	[1], [2], [3], [4], [5]	A rank-based aggregation transformer, drift-aware streaming detectors, a hyperbolic graph embedding, a scaled classical estimator, and a parallel explainer; four of the five optimize a resource rather than accuracy.
B2	Platforms, pipelines, and infrastructure	[7], [8], [9]	A three-layer stack: distributed-training economics in time, communication, and energy, a Spark cluster engine, and a Kafka and Spark Streaming fog/cloud tier. No paper connects the layers.
B3	Applications and domains	[10], [11], [12], [13], [14]	Security, telecom, finance, agriculture, and affective computing; method-building concentrates where the data has native graph or stream structure, and the “big data” label loosens elsewhere.
B4	Foundation models and large language models	[6], [15], [16], [17]	The classical-to-foundation-model arc, a multimodal industrial LLM, an LLM coupled to graph learning, and an agentic survey; none is tested under streaming or cluster constraints.
B5	Privacy, governance, and evaluation	[18], [19], [20]	Adaptive federated aggregation with a privacy–utility table, a comparative review of synthetic-data generation, and the corpus’s only empirical reproducibility audit.